FIGURE S1 | A: DESIGN BRIEF

Define the problem before selecting the configuration

Five framing concerns and a shared decision specification; an adaptation, not a published five-part taxonomy.

1 Twinning object & scope

What is represented?

Which entity, process or system?
What are its boundaries and interactions?
Which states, behaviours and scales matter?

OUTPUT Counterpart, scope and required detail. [R1, R5]

2 Purpose

Why is the twin needed?

Which problem or decision must it support?
Who uses it, and what value is expected?
What objectives and constraints define success?

OUTPUT Intended use and success criteria. [R1, R5]

3 System architecture brief

What solution is envisaged?

Which capabilities, models and components?
Which interfaces and deployment constraints?
Who analyses, selects, approves and executes?

OUTPUT Architectural intent and responsibilities. [R1, R2, R6]

4 Data requirements

What evidence and information are needed?

Which sources, meanings and quality levels?
What timing, coverage and uncertainty?
How are fitting and evaluation data managed?

OUTPUT Data needs and evidence arrangements. [R1, R3, R12]

5 Service requirements

What must the twin deliver, to whom, and within which performance and authority limits?
Monitoring | Diagnosis | Prediction | Scenario analysis | Recommendations | Permitted actions
OUTPUT Required outputs, recipients, delivery mode and acceptance conditions. [R1, R5, R6]

DECISION SPECIFICATION | SHARED ACROSS THE FIVE CONCERNS

Subject
Twin design, model credibility, or a decision about the target?

Alternatives
What may be selected or changed? Specify the decision variables.

Objectives & constraints
What is improved, and which limits must be respected?

Scope & horizon
Which entities and interactions? Over what consequence horizon?

Required evidence
Which observations, predictions and credibility evidence?

Responsibility & authority
Who analyses, chooses, approves and implements the decision?

A informs both B (configuration choices) and C (functional allocation). Design can be iterated.

The five-card arrangement and six-field decision specification are proposed synthesis.

Xiong et al. [R1] distinguish object, purpose, architecture, data, modelling and services; these are not B's six axes.

The decision subject, not system size alone, determines evidence needs and appropriate authority. [R5, R6, R13]